



MY LIGHT-UP LIGHTHOUSE!

A Fun STEM Story & Activity Book

Name: _____

The Dark and Wild Sea 🌊

Imagine you are sitting in a boat at night. The sun has gone down, and the sky is pitch black. The sea is very deep, dark, and dangerous!

How do big ships find their way home safely when they cannot see anything?

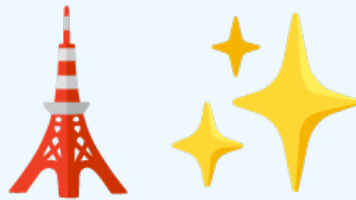


The Hero Lighthouse!



Look! A giant tower stands tall on the rocks. It is a **Lighthouse!**

It shoots a powerful, bright beam of light across the water. Coastal guards in India use these tall towers to guide boats safely through the dark waves.



Meet Your Circuit Friends

Today, you are a mini engineer! We are going to make our own glowing lighthouse. Let's meet our new electronic buddies:



LED Bulb

A tiny, baby lamp that loves to glow bright.



Coin Cell Battery

A flat, silver button full of hidden electrical power.



Copper Tape

A special, super shiny track that moves electricity!

The Train Track Secret

What is a **circuit**? Think of it like a toy train track loop!

- The **battery** is the engine that gives a strong push.
- The **copper tape** is the metal track.
- The **LED bulb** is a station that lights up when the train passes!

 **The Golden Secret:** If there is a break or gap in the track, the train stops! Electricity needs a complete, closed loop to work.

Lighthouses of India 🇮🇳

Lighthouses are real! India has beautiful ones along its coasts:

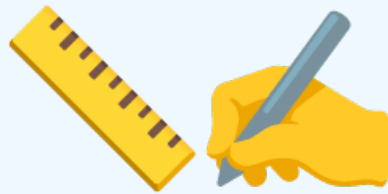
- **Dolphin's Nose:** A famous lighthouse in Visakhapatnam that sits on a giant hill shaped like a dolphin's nose!
- **Aguada Lighthouse:** A very old, historic tower in sunny Goa.
- Many are painted with bright **red-and-white stripes** so ships can see them during the day too!



Step 1: Building the Track 🎀

Let's build! Take your shiny copper tape.

- 1 Peel off the paper back carefully.
- 2 Stick the tape flat along the dotted guidelines on your paper lighthouse page.
- 3 Press down hard with your thumb to keep it perfectly smooth. No breaks allowed!



Step 2: Placing the Battery

Now, let's add the power engine!

Take your silver coin cell battery. Find the side with the little plus sign (+). That is the positive side!

Place the battery flat down on your shiny copper track, keeping the (+) side facing up to the sky.



Step 3: Connecting the Lamp



Look closely at your baby LED lamp. It has two metal legs. One leg is longer, and one leg is shorter!

- Place the **long leg** on the track connected to the plus (+) side.
- Place the **short leg** on the other side of the track.

They must touch the shiny copper tracks perfectly!

The Light-Up Magic! ✨

Are you ready? Press down gently on your battery and legs...

CHAMATKAR! ✨

The LED bulb glows bright! You have successfully completed the circuit and made electricity flow. Your lighthouse is officially alive!



The Detective Challenge

Uh-oh! Is your light not turning on? Don't worry! Put on your **Electricity Detective** hat and check these three clues:

Clue 1: Is there a crack or gap in your copper tape? Smooth it down or add a tiny patch to fix the track!

Clue 2: Is your battery upside down? Flip it so the (+) side faces up!

Clue 3: Is the LED leg switched? Flip the lamp legs around!

Decoration Time! 🎨

Now make your lighthouse unique and beautiful!

Grab your crayons, colored pencils, or stickers. Draw big splashing ocean waves around the bottom. Draw tiny wooden sailboats or jumping dolphins in the water. Add some twinkling yellow stars in the sky!



My Circuit Champion Badge

Let's check your new superpowers:

- ☐ I can find a battery, copper tape, and an LED bulb!
- ☐ I know that electricity needs a full loop to travel.
- ☐ I made my lighthouse bulb light up!
- ☐ I can name an Indian city with a lighthouse.



CONGRATULATIONS!

You are officially a Circuit Champion!



THANK YOU!

Keep your light shining bright, little engineer! See you on our next science adventure!